Process Characterization Plan

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCP-010

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Field	Value
Document ID	PCP-010
Document class	Process Characterization Plan
Title	Process Characterization Plan: Ultrafiltration / Diafiltration (Step 10)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Formulation / Drug Product Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Analytical Development	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

CPP, critical process parameter · CQA, critical quality attribute · DF, diafiltration · DP, drug product · DS, drug substance · DV, diavolume · GPP, general process parameter · ICH, International Council for Harmonisation · IPC, in-process control · KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · TFF, tangential-flow filtration · TMP, transmembrane pressure · UF, ultrafiltration · UF/DF, ultrafiltration / diafiltration.

1 Purpose and scope

This Process Characterization Plan (**PCP-010**) defines the Stage 1 (Process Design) evaluation of the A-Mab final ultrafiltration / diafiltration (UF/DF) operation (Step 10), the tangential-flow-filtration step that concentrates the virus-filtration pool and buffer-exchanges it into the final formulation buffer to deliver the drug substance at its target concentration. The plan specifies the objectives, the process-performance measures in scope, the process parameters and ranges to be evaluated, the analytical methods, the sampling plan, the analysis approach, and the acceptance and decision criteria by which each parameter will be classified.

Scope. The commercial-scale UF/DF operation that concentrates and buffer-exchanges the virus-filtration pool to the drug-substance target of 75 g/L, evaluated on a qualified bench-scale UF/DF model. The upstream virus-filtration step (Step 9), which defines the feed to this operation, is addressed only where it defines that feed; its characterization is out of scope. In the A-Mab case study the operation has **no drug-substance product-quality impact** and the **formulation characterization** (excipient levels, pH, osmolality and stability) is conducted and reported under the **drug-product development program**; that characterization is therefore out of scope of this drug-substance plan. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-010**).

Basis. In the A-Mab case study, final UF/DF is a formulation / mass-balance operation with no drug-substance product-quality impact (CMC Biotech Working Group 2009). The Stage 1 approach and report structure follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012) and the lifecycle approach of the FDA 2011 guidance (U.S. Food and Drug Administration 2011).

2 Objectives

The evaluation will:

1. confirm that UF/DF has **no impact on the drug-substance product-quality CQAs**, which pass through the operation unchanged;
2. establish the effect of the operating parameters on the process-performance measures — step yield, buffer-exchange completeness and the final DS concentration;
3. confirm that the drug substance is delivered at its target concentration and buffer composition across the operating ranges;
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter; and
5. classify each parameter for its risk of impact to CQAs and to process performance (CPP / KPP / GPP) and provide the ranges that support the Stage 2 operating conditions and the drug-substance control strategy.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-010	Process Characterization Report (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Ultrafiltration / diafiltration is the final drug-substance operation of the A-Mab train: a tangential-flow-filtration step in which ultrafiltration concentrates the virus-filtration pool and diafiltration exchanges it into the final formulation buffer, delivering the drug substance at its target concentration (CMC Biotech Working Group 2009). Unlike the CQA-forming and CQA-clearing operations it forms no new product-quality attribute; the antibody is retained by the membrane while water and small solutes permeate, so the product-quality attributes established and cleared upstream pass through unchanged. Platform knowledge from related humanized IgG1 antibodies established that a well-controlled UF/DF step concentrates and buffer-exchanges without altering the glycan, charge-variant, aggregate or impurity attributes, and that its parameters govern process performance — buffer-exchange completeness, permeate flux/process time and the final concentration. This knowledge frames the process-performance measures in scope and the prioritization of parameters for evaluation.

4.2 Quality attributes and process-performance measures

UF/DF **sets or modifies no drug-substance product-quality CQA**; the CQAs established and cleared upstream pass through the operation unchanged. The operation is therefore evaluated against the process-performance measures it governs (Table 4.1): the product (step) yield, the buffer-exchange completeness delivered by the diafiltration, and the final DS concentration. Formulation attributes (excipient concentrations, pH, osmolality) are drug-product characterization endpoints and are addressed under the drug-product program, not here.

Table 4.1: Process-performance measures for the UF/DF evaluation (no drug-substance product-quality CQA is set by this step).

Process-performance measure	What it governs	Basis of the criterion
Product (step) yield	Antibody recovery across the final operation	Mass balance; platform target
Buffer-exchange completeness	Exchange into the final formulation buffer	Number of diavolumes (Table 6.1)
Final DS concentration	Drug-substance concentration delivered	NOR 70–80 g/L (Table 6.1)
Product-quality CQAs (monitored)	Confirmation of no product-quality impact	Set and cleared upstream; unchanged here

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments, the evaluation scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). Because UF/DF carries no credible risk of impact to a drug-substance product-quality CQA, none of its parameters was assigned to a multivariate design of experiments; each parameter is evaluated **one factor at a time** across its characterization range for its effect on the process-performance measures. The resulting parameter set and study-type assignment are given in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Evaluation will be performed on a qualified bench-scale UF/DF (tangential-flow-filtration) model representing the commercial operation — the platform membrane chemistry and molecular-weight cut-off, operated at the commercial-equivalent membrane loading, transmembrane pressure and cross-flow, with the same ultrafiltration-concentration and diafiltration-exchange sequence (**SOP-2013**). Prior to evaluation the model will be qualified against at-scale reference data (**SOP-1001**) by comparing step yield, the final concentration and the buffer-exchange completeness, confirming the model is representative and suitable to support the parameter ranges (Parenteral Drug Association 2013).

5.2 Operation

The virus-filtration pool is concentrated by ultrafiltration, diafiltered against the final formulation buffer for the target number of diavolumes, and further concentrated to the final drug-substance target, with the transmembrane pressure and cross-flow controlled through the operation. The number of diavolumes, the transmembrane pressure and the final DS concentration are the operating variables; membrane lot and buffer composition are controlled to platform specifications and treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009).

5.3 Analytical methods

Product concentration and the product-quality attributes will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports.

Table 5.1: Validated analytical methods for the evaluation.

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

5.4 Sampling plan

Each run will be sampled at the final drug-substance pool for concentration and for the size- and charge-variant attributes, and the product mass across the operation will be recorded for the step-yield calculation; the diafiltration exchange will be monitored to confirm buffer-exchange completeness. Runs at the parameter set-points will be replicated to quantify the reproducibility of yield and final concentration.

5.5 Statistical and analysis methods

Because no drug-substance product-quality CQA is affected, the operation is evaluated univariately rather than by a designed experiment. Each parameter will be varied one factor at a time across its characterization range, holding the others at their set-points, and the process-performance measures — step yield, buffer-exchange completeness and final DS concentration — will be summarized against the parameter level. Reproducibility of yield and final concentration will be estimated from the replicated set-point runs. Parameter ranges will be judged acceptable where the performance measures remain within their platform targets and the product-quality CQAs are unchanged.

6 Study design

6.1 Parameters, ranges and study type

3 parameters are in scope (Table 6.1). The characterization range of each parameter ("Range studied") defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the evaluation and is reported in PCR-010.

Table 6.1: UF/DF parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Number of diavolumes	DV	8	7–10	7–9	univariate
Transmembrane pressure	psi	12	10–15	10–14	univariate
Final DS concentration	g/L	75	65–85	70–80	univariate

6.2 Univariate ranging

Each parameter will be evaluated one factor at a time across its characterization range:

- **Number of diavolumes** — evaluated across its range for its effect on buffer-exchange completeness and process time. More diavolumes lower the residual old-buffer components at the cost of longer process time; the range is bounded to assure complete exchange without unnecessary process time.
- **Transmembrane pressure** — evaluated across its range for its effect on permeate flux, process time and yield. Higher pressure raises flux and shortens process time but risks membrane fouling and concentration polarization near the top of the range.
- **Final DS concentration** — evaluated as the controlled output attribute of the operation, confirming that the drug substance is delivered on target across the operating conditions.

The product-quality CQAs will be confirmed unchanged across the ranging to demonstrate the absence of a drug-substance product-quality impact.

7 Acceptance and decision criteria

The evaluation is acceptable when:

1. the scale-down UF/DF model qualification confirms representativeness of the commercial operation (**SOP-1001**);
2. the product-quality CQAs are demonstrated to be **unchanged** across the operation and across each parameter's characterization range; and
3. step yield, buffer-exchange completeness and the final DS concentration remain within their platform targets across each parameter's normal operating range, with reproducibility at the set-point sufficient to assure drug substance of consistent concentration and buffer composition.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009): a parameter that does not significantly impact a CQA but affects process performance or consistency is a **KPP**; a parameter with no meaningful impact on CQAs or performance over a wide range is a **GPP**. No UF/DF parameter is expected to be a CPP because the operation forms no drug-substance CQA; formulation-related criticality is assessed under the drug-product program.

8 Data management and integrity

All raw data, analytical results and analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst, and the ranging results will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-010.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the ranging study; author PCR-010
Formulation / Drug Product Development	Own the formulation characterization (reported under the drug-product program)
Analytical Development	Perform concentration and size/charge-variant testing under the referenced AMV methods
Manufacturing Science	Qualify the scale-down UF/DF model; confirm representativeness
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-010**), reporting the executed ranging, the parameter classification and the confirmation that the operation has no drug-substance product-quality impact. PCR-010 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows scale-down model qualification and precedes the Stage 2 PPQ campaign.

11 Risks and assumptions

The plan assumes the scale-down UF/DF model is representative of the commercial operation (confirmed by qualification) and that the platform analytical methods are validated and in control. The principal risk is that yield, flux or buffer-exchange kinetics shift at commercial scale; this is mitigated by confirming the model at scale in Stage 2 and by carrying the KPP ranges forward with in-process monitoring of the final concentration and buffer exchange. The formulation characterization is addressed under the drug-product program.

12 Approvals

Approval of this plan authorizes execution of the evaluation described herein. Signatures are recorded in Table 1.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.

International Council for Harmonisation. 2023. *ICH Q9(R1): Quality Risk Management*. ICH.

International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.

Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned univariate ranging

Each parameter is varied one factor at a time across its characterization range (Table 6.1) with the remaining parameters held at their set-points; the process-performance measures are recorded at each level.

Table 12.1: Appendix A — planned one-factor-at-a-time ranging design.

Parameter	Set-point	Range studied	NOR	Performance measures recorded
Number of diavolumes	8	7–10	7–9	step yield, buffer-exchange completeness, final DS concentration
Transmembrane pressure	12	10–15	10–14	step yield, buffer-exchange completeness, final DS concentration
Final DS concentration	75	65–85	70–80	step yield, buffer-exchange completeness, final DS concentration